

ZOREE TMS – Database Design & Migration Rules for Claude

Core Principles

- 1 Never make ad hoc schema changes
- 2 Always use versioned migration files
- 3 Define schema before UI/API development
- 4 Protect existing production data
- 5 Optimize for safety, integrity, and scalability

Schema Design Rules

- 1 Use primary keys, foreign keys, and constraints
- 2 Normalize data by default; justify denormalization
- 3 Avoid unnecessary JSON usage for relational data
- 4 Design for multi-tenant isolation
- 5 Include audit fields (created_at, updated_at, etc.)
- 6 Use consistent naming conventions

Migration Rules

- 1 Use forward-only migrations
- 2 Never drop/rename fields without safe transition
- 3 Provide rollback or fallback strategy
- 4 Ensure migrations are deployment-safe
- 5 Document all schema changes clearly

Data Integrity Rules

- 1 Enforce validation at DB level where possible
- 2 Avoid duplicate sources of truth
- 3 Define transaction boundaries for multi-step operations
- 4 Use controlled values for statuses (enums or lookup tables)

Performance & Indexing

- 1 Add indexes based on query patterns
- 2 Avoid over-indexing
- 3 Optimize for reporting and filtering needs

Operational Rules

- 1 Separate seed/demo data from production logic
- 2 Document table purpose and ownership
- 3 Ensure all changes include impact analysis (API/UI/services)

Required Output for DB Changes

- 1 Schema changes
- 2 Migration files
- 3 Backfill/data migration needs
- 4 Index changes
- 5 Constraint changes
- 6 Rollback considerations
- 7 Affected APIs/services/UI
- 8 Risks and assumptions